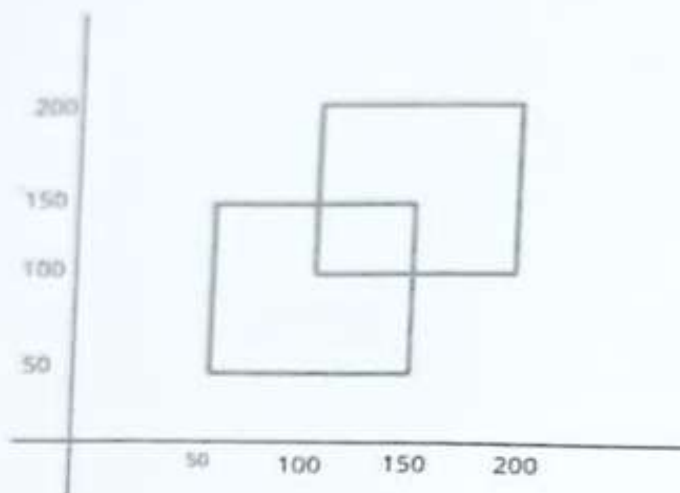


4. Given two rectangles R1 and R2, print the coordinates of clipped rects.

R1 = {(50, 50), (150, 50), (150, 150), (50, 150)}

R2 = {(100, 100), (200, 100), (200, 200), (100, 200)}

Output: {(100,100), (150, 100), (150,150), (100, 150)}



5. For a given matrix, find the maximum product of 'k' elements. The elements can be formed from continuous 4 elements horizontally, vertically or diagonally. Eg: For k= 4, the maximum product is (6*4*7*9) from the last column,

1 2 0 1 4

3 1 2 4 6

0 2 3 1 4

1 3 2 0 7

2 1 3 2 9

1	2	0	1	4
3	1	2	4	9
0	2	3	8	4
1	3	7	0	7
2	1	3	2	9

1. If $a=1, b=2, c=3, \dots, z=26$. Given a string, find all possible codes that string can generate. Give a count as well as print the strings.

For example:

Input: "1123". You need to generate all valid alphabet codes from this string.

Output List:

aabc // $a = 1, a = 1, b = 2, c = 3$

kbc // since k is 11, $b = 2, c = 3$

alc // $a = 1, l = 12, c = 3$

aaw // $a = 1, a = 1, w = 23$

kw // $k = 11, w = 23$

2. Given an unsorted array of inputs, sort the shortest sub-array which results in sorting of the whole array.

Example:

Input: [-1,0,4,3,2,1,7,8,9]

By sorting sub array [4,3,2,1] the whole Array is sorted.

Input: [10, 15, 20, 30, 25, 40, 35, 45, 50, 60]

By sorting sub array [30, 25, 40, 35] the whole Array is sorted.

Input: [-1,0,4,3,2,1,7,8,9,-2]

Shortest sub-array to be sorted is [-1,0,4,3,2,1,7,8,9,-2]

3. Check if a non-empty string can be constructed by repeating a substring multiple times.

Example:

Input: "abab"

Output: true

Input: "aba"

Output: false

Input: "abcabcabcabc"

Output: true

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General instructions

1. Read all the questions and understand them clearly before you start writing the programs. In case of any doubts, get them clarified with your invigilator.
2. Efficient and optimum logic, coding standards and modularity and will be considered for evaluation.
3. Verify the output of the programs for multiple test cases, before showing it to the invigilator.

1. Mini regex finder

Find the first match of a pattern in a given string

A pattern can have two types of special characters * and + with the following meaning

* - the previous character in the pattern can occur any number of times or not occur at all

+ - the previous character can appear any number of times and must appear at least once

Example**Input 1**

abcbbcbabb
cb*cab+

Output 1

cbbcbabb

Input 2

abcfbbbacbk
bbk*ac+

Output 2

bbac

2. Rotate a Square matrix (N x N) by 90 degrees clockwise without using a third matrix**Input:**

1 2 3

4 5 6

7 8 9

Output:

7 4 1

8 5 2

9 6 3

Input:

1 2

3 4

Output:

3 1

4 2

order while maintaining their original positions.

Example 2:

Input: nums = [4, 5, 6, 7, 0, 1, 2]

Output: nums = [0, 7, 2, 5, 4, 1, 6]

Explanation: Even numbers are sorted in ascending order, and odd numbers are sorted in descending order while maintaining their original positions.

Example 3:

Input: nums = [11, 13, 15, 17]

Output: nums = [17, 15, 13, 11]

Explanation: Odd numbers are sorted in descending order at their initial positions.

6. You have two version numbers, version1 and version2. Version numbers consist of parts separated by dots (.). Each part may have leading zeros and contains at least one character. Compare the parts as integers, ignoring leading zeros. If a version number doesn't have a part at a position, treat it as 0.

Both version1 and version2 only contain numbers and dots (.).

Both version1 and version2 are valid version numbers.

- If version1 is less than version2, return -1.
- If version1 is greater than version2, return 1.
- Otherwise, return 0.

Example 1:

Input: version1 = "1.01", version2 = "1.001"

Output: 0

Explanation: Ignoring leading zeroes, both "01" and "001" represent the same integer "1".

2. Input: version1 = "1.0", version2 = "1.0.0"

Output: 0

Explanation: version1 does not specify revision 2, which means it is treated as "0".

3. Input: version1 = "0.1", version2 = "1.1"

Output: -1

Explanation: version1's revision 0 is "0", while version2's revision 0 is "1". $0 < 1$, so $\text{version1} < \text{version2}$.

Test Case 3:

Input:

s = a

t = aa

Output:

(empty string)

Q5. Trapping Rain Water

You are given n non-negative integers representing heights of bars. Compute how much water can be trapped after raining.

Sample Test Cases:

Test Case 1:

Input:

12

0 1 0 2 1 0 1 3 2 1 2 1

Output:

6

Test Case 2:

Input:

6

4 2 0 3 2 5

Output:

9

Test Case 3:

Input:

5

2 0 2 0 2

Output:

4

Q1. Spiral Matrix Traversal

You are given a 2D matrix of size $m \times n$. Print all elements of the matrix in spiral order, starting from the top-left element.

Sample Test Cases:

Test Case 1:

Input:

3 3

1 2 3

4 5 6

7 8 9

Output:

1 2 3 6 9 8 7 4 5

Test Case 2:

Input:

4 3

1 2 3

4 5 6

7 8 9

10 11 12

Output:

1 2 3 6 9 12 11 10 7 4 5 8

Test Case 3:

Input:

2 2

1 2

3 4

Output:

1 2 4 3

Q2. Maximum Sum Submatrix (2D Kadane's Algorithm)

You are given a 2D integer matrix. Find the submatrix which has the maximum sum.

Sample Test Cases:

Test Case 1:

Input:

3 3

-1 -2 -3

-4 5 6

-7 8 9

Output:

28

Test Case 2:

Input:

2 4

1 2 -1 4

-2 3 0 5

Output:

12

Question 3:

Given a string s , find the length of the longest substring without repeating characters.

- Example Input: $s = \text{"abcabcbb"}$
- Output: 3 (The substring is "abc".)

Question 4:

Given an $m \times n$ integer matrix, if an element is 0, set its entire row and column to 0. You must do it in place.

- Example Input: matrix = $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
- Output: $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

Question 1 : Climbing Stairs

You are climbing a staircase. It takes n steps to reach the top. Each time, you can either climb 1 or 2 steps. How many distinct ways can you climb to the top?

Example 1:

Input: $n = 2$

Output: 2

Explanation: There are two ways to climb to the top.

1. 1 step + 1 step
2. 2 steps

Input: $n = 3$

Output: 3

Explanation: There are three ways to climb to the top.

1. 1 step + 1 step + 1 step
2. 1 step + 2 steps
3. 2 steps + 1 step

Constraints:

- $1 \leq n \leq 45$

Question 2 : Ordering Colors

Given an array `nums` with n objects colored red, white, or blue, sort them in-place so that objects of the same color are adjacent, with the colors in the order red, white, and blue.

We will use the integers 0, 1, and 2 to represent the color red, white, and blue, respectively. You must solve this problem without using the library's sort function.

Example 1: Input: `nums = [2,0,2,1,1,0]`

Output: `[0,0,1,1,2,2]`

Test Case 3:

Input:

3 2

-1 2

3 -2

4 1

Output:

6

Q3. Longest Palindromic Substring

Given a string *s*, return the longest substring which is a palindrome.

Sample Test Cases:

Test Case 1:

Input:

babad

Output:

bab (or aba)

Test Case 2:

Input:

cbbd

Output:

bb

Test Case 3:

Input:

a

Output:

a

Q4. Minimum Window Substring

Given two strings *s* and *t*, return the smallest substring in *s* which contains all characters of *t*. If such substring does not exist, return an empty string.

Sample Test Cases:

Test Case 1:

Input:

s = ADOBECODEBANC

t = ABC

Output:

BANC

Test Case 2:

Input:

s = a

t = a

Output:

a